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MILITARY SPECIFICATION
PROJECTILE, 4.2 INCH, W.P. SMOKE, M328A1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Projectile, 4.2 Inch, W.P. Smoke, M328A1, Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-A-2550 Ammunition and Special Weapons, General Specification for

STANDARDS
Federal
Federal Test Method Standard No. 141 - Paint, Varnish, Lacquer and Related Materials, Method of Inspection, Sampling and Testing.

Federal Test Method Standard No. 151 - Metals; Test Methods

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1315

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DRAWINGS
Munitions Command
D10534915 - Projectile, 4.2 Inch, W.P. Smoke, M328A1, Metal Parts Assembly
IEL-10534915- Inspection Equipment List for Projectile, 4.2 Inch, W.P. Smoke, M328A1, Metal Parts Assembly
F8848553 - Pallet

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents from a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA-B46.1 - Surface Texture, Surface Roughness, Waviness and Lay.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Heat Treatment or Stress Relief.-Heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable drawing.

3.2.1 Hot Forge - Heat Treatment Method.-Prior to finish machining, projectile bodies or bases manufactured from forgings shall be subjected to heat treatment to attain the required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit ($^{\circ}$F), followed by an oil quench. After quenching, the forgings shall be tempered without delay at a temperature above 750° F. suitable to produce required mechanical properties. After removal from the tempering furnace, the forging shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700°F., after which cooling may be accelerated by water spray.

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3.2.1.1 Bases (Alternative Design).-Bases may be heat treated by normalizing in lieu of quenching and tempering to obtain specified mechanical properties. The bases shall be heated to a temperature of 1600°F., minimum, held at that temperature for a minimum time of 30 minutes, and cooled in such a manner as to produce uniform mechanical properties throughout the component.

3.2.2 Cold Shaping and Hot Cup - Cold Draw Method.-Cold working shall be so accomplished as to produce in the projectile body or base the mechanical properties specified. The projectile body or base shall be stress relieved at a temperature of 700°F., minimum, following the final forming operation during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be established which will produce mechanical properties specified on the applicable drawing. The temperature and time duration of a process anneal prior to the final forming operation shall be established by the ability to meet the required mechanical properties when formed by an approved sequence of operations.

3.2.3 Tubing.-Bodies (alternative design) manufactured from tubing shall be stress relieved after cold forming of the nose.

3.3 Mechanical Properties.-Mechanical properties as specified on the applicable component drawing shall apply throughout the component.

3.4 Quality of Joint.-The total area of inclusions, lack of adhesion, solder groove or other defects shall not exceed 25 percent of the total faying surfaces of the joint.

3.5 Functional.

3.5.1 Excess Pressure.-When tested in accordance with 4.4.6.1 the projectile assembly shall reveal no evidence of metal parts failure. Metal parts failure is defined as: "Projectile body or base deformation or separation of components in the mortar tube or in flight attributable to noncompliance with drawing or specification requirements".

3.6 Workmanship.-The requirements for workmanship are as shown on the applicable drawing in the applicable specifications and the following:

3.6.1 Metal Defects.-All components shall be free from cracks, splits, cold shuts, pipe, porosity, inclusions or any similar defect.

3.6.2 Surface Finish.-The requirements for surface finish are as detailed on the applicable drawing. Each body cavity shall be free of scale, fins, deep seams, die scratches, pits or other surface discontinuities.

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3.6.3 Threads.-Thread(s) shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.6.4 Painting.-Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before testing and packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.6.5 Burr.-No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.6.6 Foreign Matter.-No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.-Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.2 First Article Sample

4.2.1 Initial Production Sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 43 metal parts assemblies, as follows:

30 Metal Parts Assemblies complete including paint
5 Metal Parts Assemblies complete except paint
5 Sets component parts unassembled and unpainted
3 Body and vane assemblies unpainted

Twenty-five of the complete and painted metal parts assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and Test.-After inspection and provisional acceptance at source, the sample will be inspected for all the requirements of the drawings and specifications at a Government Laboratory or such other facility specified in the contract.

4.2.1.2 Initial Production Sample Failure.-Failure of the sample to comply with the requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection Provisions

4.3.1 Lot

4.3.1.1 Submission of Product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Heat Treatment Lots.-Heat treatment lots shall contain body or base forgings from not more than one heat of steel that have been heat treated in the same heat treating equipment, at the same temperature, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Stress Relieved Lots.-Stress relieved lots shall contain cold worked bodies or bases from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4 Tubing lots.-Bodies manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. The bodies shall be cold nosed and stress relieved using the same equipment and procedures. Lots of bodies thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.4.1 Adapters.-Adapters manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. Tubing lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.5 Bar Stock Lots.-Bar stock lots shall contain not more than one heat of steel. Bar stock lots thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

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4.3.1.6 Miscellaneous Lots.-Small lots (such as bodies manufactured from abnormally small heats of steel or bodies and bases that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or tubing lots as the case may be. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.2, 4.3.1.3 or 4.3.1.4 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.7 Projectile Lots.-When it has been determined that a heat treated, stress relieved, tubing, bar stock or miscellaneous lot of bodies or bases have complied with the mechanical properties requirement of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number in accordance with MIL-STD-1168. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day or work week. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. Continuous sampling in accordance with MIL-STD-1235 may be used if approved by the procuring activity. When performing pressure tests, the item being tested will be observed for non-compliance during the entire time it is required to be pressurized. All non-conforming material shall be rejected. The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis for surface roughness determination.

4.3.2.1 Control Tool Manufacture.-When component drawings designate certain dimensions"*control tool manufacture' two components shall be selected at the start of each shift or whenever a tooling change or modification takes place, the dimensions developed by the tooling shall be gaged or measured to determine compliance with the applicable drawing. When a tooling change or modification takes place, production will not be resumed until it has been determined that developed dimensions are within specified tolerances.

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4.3.2.1 (Cont'd)

Failure of any component to comply with the requirements shall be cause for rejection of the component. Components that have been processed since the last satisfactory inspection need not be reinspected or screened, but immediate corrective action shall be taken to eliminate the discrepancy in future production.

4.3.2.2 Adapter, Nose Drawing D8798377, covering a detail of drawings F8797841 and D10534915

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Diameter of braze surface	Gage
MINOR	
201. Diameter of shoulder	Gage
202. Diameter, small outside, rear of taper	Gage
203. Length from front face to outside diameter rear of taper	Gage
204. Length from front face to shoulder	Gage
205. Length from front face to rear of braze surface	Gage
206. Length, total	Gage
207. Dimensions and location of solder ring groove incorrect (if applicable)	Visual (see 6.1)
208. Surface finish, improper	Visual (see 6.1)
209. Radii or chamfers missing or incorrect	Visual (see 6.1)
210. Burr	Visual
211. Foreign matter	Visual

4.3.2.3 Body, Drawing D8800091 and body (alternative design) drawing D8800090, covering a detail of drawings F8797841 and D10534915

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Braze surface at rear not functionally concentric with inside diameter (if applicable)	Gage

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4.3.2.3 (Cont'd)

Defect		Method of Inspection
MAJOR		
102.	Diameter, inside rear	Gage
103.	Diameter, inside front (if applicable)	Gage
104.	Diameter of braze surface at rear (if applicable)	Gage
105.	Depth of braze surface at rear (if applicable)	Gage
MINOR		
201.	Angle inside incorrect	Gage
202.	Depth from angle (inside)	Gage
203.	Surface finish, improper	Visual (see 6.1)
204.	Radii or chamfers, missing or incorrect	Visual (see 6.1)
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.2.4 Vane, drawing F879843 covering a detail of drawings F8797841 and D10534915

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Length, total	Gage
202.	Radius in center	Gage

4.3.2.5 Base, (alternative design) drawing D10534167, covering a detail of drawing F8797841

Defect		Method of Inspection
CRITICAL		
1.	Metal defective	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter of flange	Gage
102. Diameter of braze surface	Gage
103. Thickness through bottom of cavity including rotating disc seat	Gage
104. Length, total	Gage
MINOR	
201. Thickness through flange including rotating disc seat	Gage
202. Dimension or location of solder ring groove (if applicable)	Visual (see 6.1)
203. Depth of undercut, max. (if applicable)	Visual (see 6.1)
204. Width of undercut, max. (if applicable)	Visual (see 6.1)
205. Surface finish, improper	Visual (see 6.1)
206. Radii or chamfers, missing or incorrect	Visual (see 6.1)
207. Burr	Visual
208. Foreign matter	Visual

4.3.2.6 Body and vane assembly, drawing F10534916, and covering a detail of drawings F8797841 and D10534915

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter at front face	Gage
102. Diameter of braze surface of nose	Gage
103. Depth from front face of body to front face of vane, min.	Gage
104. Thickness through base including rotating disc seat	Gage
105. Thickness of wall at nose	Gage
106. Thickness of wall at datum length, min.	Gage
107. Variation in wall thickness at nose, max.	Gage
108. Variation in thickness of body wall, max.	Gage
109. Weld(s) missing or improper	Visual

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Length from front face to rotating disc seat	Gage
202. Length total, max.	Gage
203. Surface finish, improper	Visual (see 6.1)
204. Radii or chamfers, missing or incorrect	Visual (see 6.1)
205. Burr	Visual
206. Foreign matter	Visual

4.3.2.7 Body assembly, drawing F8797841 covering a detail of drawing D10534915

Defect	Method of Inspection
CRITICAL	
1. Failure to pass hydrostatic test (as evidenced by absence of stamp)	Visual
2. Failure to pass air test (as evidenced by absence of stamp)	Visual
MAJOR	
101. Weight	Approved scale
102. Concentricity of pitch diameter of fuze thread with bourrelets, including perpendicularity of front face	Gage
103. Concentricity of press fit surface counterbore with bourrelets	Gage
104. Concentricity of press fit surface with bourrelets	Gage
105. Concentricity of ogive with bourrelets	Gage
106. Concentricity of body with bourrelets	Gage
107. Concentricity of base with bourrelets (if applicable)	Gage
108. Concentricity of pitch diameter of base thread with bourrelets, including perpendicularity of disc seat	Gage
109. Vane cavity not functionally concentric with press fit surface counterbore	Gage
110. Symmetry of rotating disc seat with major diameter of base thread	Gage

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hereinafter prescribed of samples for the tension test

4.3.2.7 (Cont'd)

Defect	Method of Inspection
MAJOR	
111. Diameter of bourrelets, min.	Gage
112. Diameter of body	Gage
113. Length total, max.	Gage
114. Improper assembly of adapter	Gage
MINOR	
201. Diameter of front face of adapter	Gage
202. Diameter of counterbore forward of fuze thread	Gage
203. Diameter of undercut rear of fuze thread	Gage
204. Diameter of "O" Ring groove	Gage
205. Diameter of press fit surface counterbore	Gage
206. Diameter of base thread undercut	Gage
207. Location of forward basic diameter	Gage
208. Length to rear of forward bourrelet	Gage
209. Length to front of rear bourrelet	Gage
210. Length of rotating disc seat	Gage
211. Length of thread base	Gage
212. Depth of counterbore forward of fuze thread	Gage
213. Width of undercut rear of fuze thread	Gage
214. Location of "O" Ring groove	Gage
215. Depth of fuze thread cavity	Gage
216. Width of "O" Ring groove	Gage
217. Width of press fit surface counterbore	Gage
218. Width of base thread undercut	Gage
219. Width across rotating disc seat	Gage
220. Staking notches missing or incorrect	Visual (see 6.1)
221. Radii or chamfers, missing or incorrect	Visual (see 6.1)
222. Marking missing, incorrect or illegible	Visual
223. Burr	Visual
224. Foreign matter	Visual

4.3.2.8 Metal parts assembly drawing D10554915

Defect	Method of Inspection
CRITICAL	
1. Diameter of press fit surface	Gage
2. Presence of reverse taper in press fit surface	Gage
3. Radius not tangent with press fit surface for complete circumference	Visual/Gage (see 6.1)
4. Surface finish of press fit surface	Visual/Gage
5. Paint on press fit surface	Visual

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of fuze thread, max.	Gage
102. Minor diameter of fuze thread, max.	Gage
103. Pitch diameter of base thread, min.	Gage
104. Major diameter of base thread, min.	Gage
105. Diameter of bourrelets, max.	Gage
106. Length of effective fuze thread, min.	Gage
107. Corrosive preventive compound missing from press fit surface	Visual
MINOR	
201. Pitch diameter of fuze thread, min.	Gage
202. Pitch diameter of base thread, max.	Gage
203. Corrosive preventive compound missing from threads	Visual
204. Protective coating, inadequate or damaged	Visual
205. Thread(s) damaged or not full	Visual
206. Marking missing, incorrect or illegible	Visual
207. Foreign matter	Visual
208. Protector cap(s) missing	Visual
209. VCI paper missing	Visual

4.3.2.9 Pallet, drawing F8848553

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Pallet damaged (unserviceable)	Visual
202. Sleeve missing	Visual
203. Improperly packed	Visual
204. Strapping missing, loose or broken	Visual and Manual
205. Marking missing, incorrect or illegible	Visual

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4.3.3 Inspection Testing. - The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Mechanical Properties

4.3.3.1.1 Tension Test

4.3.3.1.1.1 Body. - Two bodies from each heat treatment, stress relieved, tubing or miscellaneous lot as defined by 4.3.1.2, 4.3.1.3, 4.3.1.4 and 4.3.1.6 as applicable, after final heat treatment or stress relief shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of one or more tension test specimen to comply shall be cause for rejection of the lot of bodies represented by the failing specimen.

4.3.3.1.1.2 Base. - Two bases from each heat treatment or stress relieved lot as defined by 4.3.1.2, 4.3.1.3 or 4.3.1.6 as applicable shall be subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of either tension test specimen to comply shall be cause for rejection of the lot of bases represented by the failing specimen.

4.3.3.1.1.3 Adapters. - Two adapters shall be selected from each lot and subjected to this test to determine compliance with the mechanical properties requirements of the applicable drawing. Failure of either tension test specimen to comply shall be cause for rejection of the lot.

4.3.3.1.1.3.1 Machined From Tubing. - With the approval of the procuring activity, when the adapter is machined from tubing in such a manner that the mechanical properties are not altered, test coupons approximately 6 inches in length may be removed from one end of each of 3 tubes to represent the heat lot and tested in lieu of testing the component. Failure of one or more tension test specimen to comply with the yield strength and elongation requirements of the applicable component drawing shall be cause for rejection of the tubing lot.

4.3.3.1.1.4 Machined From Bar Stock. - With the approval of the procuring activity when the base or adapter is machined from bar stock in such a manner that the mechanical properties are not altered, test coupons approximately 6 inches in length may be removed from one end of each of three bars representing the heat lot and tested in lieu of testing the component. Failure of one or more tension test specimen to comply with the yield strength and elongation requirements of the applicable component drawing shall be cause for rejection of the bar stock lot.

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4.3.3.2 Torque Test. - (Major Defect) Each body and vane assembly after spot welding and nosing but prior to assembling nose adapter, shall be subjected to this test to determine compliance with the requirements of the applicable drawing. Movement of the vane shall be cause for rejection of the assembly.

4.3.3.3 Brazed Joint

4.3.3.3.1 Hydrostatic Test. - (A Critical Defect) - Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.3.2 Air Pressure Test. - (A Critical Defect) - Each metal parts assembly shall be subjected to this test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly.

4.3.3.3.3 Quality Of Joint. - Two body assemblies prior to painting shall be selected from each lot and subjected to this test to determine compliance with 3.4. Failure of any segment of either joint to comply shall be cause for rejection of the lot.

4.3.3.4 Cleaning and Application of Protective Coating. - The sample size and acceptance criteria for the inspection of cleaning the item and application of protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.4 Functional. - The following proving ground tests will be performed by the Government.

4.3.4.1 Excess Pressure. - Twenty-five metal parts assemblies from the initial production sample will be ballistically tested to determine compliance with 3.5.1. Failure of one or more assemblies to comply will be cause for disapproval of the initial production sample.

4.3.5 Inspection Equipment.- Inspection Equipment List IEL 10534915 identifies the inspection equipment required to perform the examinations and tests prescribed herein.

4.4 Test Methods And Procedures

4.4.1 Mechanical Properties

4.4.1.1 Tension Test

4.4.1.1.1 Body. - From the bodies selected in accordance with 4.3.3.1.1.1 for the convenience of the Government one tension test specimen conforming to the standard "T" type specimen as prescribed by Federal Test Method Standard No. 151

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4.4.1.1.1 (Cont'd)

shall be taken longitudinally from each side of each body 180 degrees apart and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.1.2 Base. - From each base selected in accordance with 4.3.3.1.1.2 one tension test specimen conforming to the largest obtainable standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from the center of the base in a transverse direction normal to the centerline of the part and tested in accordance with the criteria for determinign yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method, the limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.1.3 Adapters. - From each adapter selected in accordance with 4.3.3.1.1.3, one tension test specimen conforming to the largest obtainable "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from the adapter, and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.1.3.1 Machined From Tubing. - From the test coupons supplied in accordance with 4.3.3.1.1.3.1 one tension test specimen conforming to the largest obtainable standard "R" type tension test specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each test coupon and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method, the limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.1.1.3.2 Machined From Bar Stock. - From the samples supplied by 4.3.3.1.1.4 one type "R" tension test specimen shall be taken longitudinally from midradius of each coupon, and tested in accordance with the criteria for determining yield strength and elongation as prescribed by Federal Test Method Standard No. 151. Yield strength shall be determined by the offset method, the limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.4.2 Control Tool Manufacture. - The components furnished in accordance with 4.3.2.1 shall have all dimensions gaged or measured to determine compliance with drawing requirements.

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4.4.3 Torque Test. - The body and vane assembly shall be mounted in an approved holding device. The torque shall be applied gradually for the full length of the vane until the required torque is reached.

4.4.4 Brazed Joint

4.4.4.1 Hydrostatic Pressure. - This test will be performed using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure has been reached, after which the pressure will be maintained for the minimum prescribed time. The equipment shall be calibrated prior to the start of each day's operation, and at intervals of not more than 4 hours of operation. The equipment shall also be calibrated prior to restarting the operation after a shut down of one hour or more. After acceptance, stamp the letter "H" as indicated on assembly drawing.

4.4.4.2 Air Pressure. - This test will be performed after the hydrostatic test and drying cycle as specified by the applicable drawing, using equipment capable of performing this test accurately and consistently. The assembly shall be submerged in water in such manner that the base (alternative design) and nose adapter joint(s) are covered. Observation shall be made 360 degrees of the periphery for the entire time to be pressurized, for air bubbles rising to the surface of the water or appearing in the vicinity of the base or adapter joint(s). In performing the test the pressure will be applied until the required minimum pressure is reached, after which the pressure will be held for the minimum length of time specified on the applicable drawing. The equipment shall be calibrated prior to the start of each day's operation and after each 4 hours continuous operation. After acceptance stamp letter "A" as indicated on the assembly drawing.

4.4.4.3 Sectionalized Specimens. - The projectile assemblies selected in accordance with 4.3.3.3.3 shall be sectionalized by cutting through the body at least two inches to the rear of the brazed area at the forward joint and at least two inches to the front of the brazed area of the rear joint. Each of the sections containing the brazed joints shall be cut into two 180 degree segments.

4.4.4.3.1 Quality Of Joint. - The sidewall shall be peeled or stripped from each segment by applying continuous pressure. The braze surface of each segment of the base (alternative design) and the adapter and each segment of the respective side wall shall be visually examined to determine compliance with the requirements of 3.4.

4.4.5 Protective Coating. - The procedures for testing the cleaning of components and assemblies and the application of protective coating shall be in accordance with the applicable specifications. The salt spray test as specified by the assembly drawing shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141.

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4.4.6 Functional

4.4.6.1 Excess Pressure. - The metal parts assemblies provided by 4.3.4.1 inert loaded to the service weight will be dummy fuzed and fired in a mortar for which the round is standard. The propellant charge will be adjusted to obtain 112 percent of the maximum rated pressure of the round in a new gun. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile. An attempt will be made to recover at least 60 percent of the projectiles fired. The recovered projectile will be examined to determine if deformation occurred in the mortar as evidenced by the presence of rifling marks extending more than 180 degrees around the circumference, or by an increase in body diameter in the vicinity of the bourrelets. (Beyond drawing tolerances) dishing of the base or other evidence of component failure due to firing stresses."

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level C. - Metal parts assemblies shall be palletized in accordance with drawing F8848553 and MIL-STD-1169.

5.2 Data Card. - Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Visual Examination Qualification. - When compliance with the applicable requirements are in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.2 Ordering Data. - Procurement documents should specify the title, number, and date of this specification.

6.3 Supersession Data. This specification includes the requirements of Purchase Description FA-PD-MI-2666, Rev. 0 dated 19 May 1965.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1315-A528

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

( ) DIRECT GOVERNMENT CONTRACT ( ) SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

( ) YES ( ) NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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